

Healthcare Patient Records Analytics

Business Problem Statement | Massachusetts General Hospital Synthetic Dataset

Project Type	End-to-End Healthcare Analytics Portfolio Project
Dataset	Synthetic data on approximately 1,000 patients of Massachusetts General Hospital, 2011-2022
Tools	Excel, Python, SQL Server, Power BI
Document	Problem Statement PDF

1. Background

Massachusetts General Hospital is represented here through a synthetic patient-records dataset covering the period 2011-2022. The data includes patient demographics, payer and insurance coverage, medical encounters, procedures, claim costs, payer contribution, patient responsibility, and repeat-visit behavior.

The project simulates a real-world hospital analytics requirement: turn fragmented operational and financial data into a decision-ready view for hospital management, finance teams, insurance teams, and care-coordination stakeholders.

2. Business Problem

Hospital leaders need a clear view of patient volume, hospital load, insurance coverage, procedure workload, treatment cost, and repeat-visit behavior. Without an integrated analytics layer, it is difficult to identify high-load periods, high-cost procedures, uninsured financial burden, and high-utilization patients who may require focused care coordination.

The business challenge is to create an end-to-end analytical system that converts raw patient records into actionable insights using Excel, Python, SQL Server, and Power BI.

3. Core Business Questions

- How many patients have been admitted or readmitted over time?
- How long are patients staying in the hospital, on average?
- How much is the average cost per visit?
- How many procedures are covered by insurance?

4. Expanded Real-World Analytics Questions

No.	Business Question	Stakeholder	Expected Output
1	How many total encounters occurred each year?	Hospital Management	Yearly hospital load trend
2	For each year, what percentage of encounters belonged to each encounter class?	Operations Team	Encounter mix by ambulatory, outpatient, emergency, urgent care, inpatient, and wellness
3	What percentage of encounters were over 24 hours versus 24 hours or less?	Operations Team	Short vs long-duration hospital visits
4	How many encounters had zero payer coverage, and what percentage of total encounters does this represent?	Finance Team	Uninsured or zero-coverage exposure
5	What are the top 10 most frequent procedures and the average base cost for each?	Clinical Operations	High-volume procedure workload
6	What are the top 10 procedures with the highest average base cost and how often were they performed?	Finance + Clinical Leads	High-cost procedure monitoring
7	What is the average total claim cost for encounters, broken down by payer?	Insurance Team	Payer-wise claim-cost comparison
8	How many unique patients were admitted each quarter over time?	Hospital Management	Quarterly patient demand trend
9	How many patients were readmitted or repeated within 30 days of a previous encounter?	Care Coordination Team	30-day repeat encounter rate

No.	Business Question	Stakeholder	Expected Output
10	Which patients had the most repeat visits or readmissions?	Care Coordination Team	High-utilization patient identification

5. Project Objectives

- Build an end-to-end analytics workflow from raw CSV files to interactive dashboard storytelling.
- Use Excel for initial exploration, column profiling, and basic pivot-based understanding.
- Use Python for data cleaning, date formatting, feature engineering, exploratory analysis, and cleaned CSV exports.
- Use SQL Server for table creation, import validation, business queries, and Power BI reporting views.
- Use Power BI to create an interactive dashboard with KPIs, slicers, navigation, drillable visuals, and insight summaries.
- Generate stakeholder-ready insights for hospital operations, finance, insurance coverage, procedure workload, and repeat-visit behavior.

6. Stakeholders

Stakeholder	Business Need
Hospital Management	Needs patient volume trends, peak load periods, encounter class distribution, and operational pressure points.
Finance Team	Needs claim cost, payer contribution, patient responsibility, and zero-coverage exposure.
Insurance / Payer Team	Needs payer-wise coverage ratio, average claim cost, and high-risk payer categories.
Clinical Operations	Needs most common procedures, highest-cost procedures, and procedure workload trends.
Care Coordination Team	Needs high-utilization patients, 30-day repeat encounters, and demographic repeat-visit patterns.

7. Analytical Deliverables

- Cleaned datasets and summary CSV files
- Excel column profile and pivot-based initial analysis
- Python cleaning and EDA notebooks
- SQL Server validation queries, business queries, and reporting views
- Power BI dashboard with Home, Operations, Cost & Coverage, Procedures, and Readmissions pages
- Business insights and recommendations for GitHub and LinkedIn portfolio presentation

8. Expected Business Value

The analysis helps hospital stakeholders monitor patient flow, understand cost and coverage gaps, identify high-cost procedures, evaluate payer risk, and detect high-utilization patients. The final dashboard converts healthcare records into a practical decision-support tool for operational, financial, and care-management planning.